

Product Requirements Document (PRD)

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Product: Cubyntra

Organization: Necookie Labs

Tagline: See it. Solve it.

Document Version: 1.0

Status: V1 Release

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1. Product Overview

Cubyntra is a high-precision, web-based Computer Vision Rubik's Cube scanner, state validator, solver, and interactive 3D digital twin. Developed by Necookie Labs, Cubyntra enables any user with a standard camera-enabled device (smartphone, laptop, or desktop webcam) to scan a scrambled physical 3×3 Rubik's Cube, receive immediate mathematical validation, calculate an efficient solution sequence using Herbert Kociemba's two-phase algorithm, and follow intuitive 3D animated step-by-step guidance.

Cubyntra runs 100% on the client side: no video streams or image data ever leave the user's browser, guaranteeing absolute privacy.

2. Product Vision

To create the most seamless, privacy-respecting, and mathematically rigorous physical-to-digital puzzle solving experience on the web. Cubyntra bridges physical reality with digital twin computation, laying the architectural groundwork for future autonomous physical robotics.

3. Problem Statement

The standard 3 x 3 x 3 Rubik's Cube has 43,252,003,274,489,856,000 possible permutations. While deterministic algorithms exist to solve any valid state, everyday users face major friction points:

- Manual Color Entry is Tedious & Error-Prone: Entering 54 individual facelet colors manually takes 3–5 minutes and frequently suffers from typos that make the cube unsolvable.
- Cloud-Based Solvers Violate Privacy: Existing scanning tools upload webcam frames to external servers or cloud vision APIs.
- Abstract Move Notation Confuses Beginners: Standard Singmaster notation (R, U', F2) lacks intuitive visual orientation cues, leading users to execute wrong turns.
- Physical Cube Errors Go Unexplained: If a cube has an internally twisted corner, flipped edge, or swapped stickers, typical solvers simply crash with generic "Invalid Cube" messages without diagnosing what is wrong.

4. Target Users

- Novice Puzzle Solvers: Individuals who scrambled their cube and need clear, step-by-step 3D visual guidance to restore it.
- Speedcubers & Enthusiasts: Cubers looking for efficient move sequences (HTM metric) and reconstruction diagnostics.

- STEM Educators & Students: Learners exploring group theory, computer vision, and coordinate geometry.
 - Robotics Engineers: Developers seeking a solid algorithmic and perceptual foundation for physical cube manipulation mechanisms.
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5. User Needs

- Fast, guided scanning that does not require tedious manual input.
 - Real-time feedback on whether a face is aligned and stable.
 - Complete privacy protection (no video uploads).
 - Clear, unambiguous 3D layer animations and directional arrows showing exactly how to turn each layer.
 - Granular error diagnostics explaining specifically which face or piece has an issue.
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6. Product Goals

- Deliver sub-second physical cube state reconstruction from a guided 6-face sequence.
 - Execute Kociemba's two-phase algorithm client-side, returning solutions averaging 18–22 moves.
 - Render an interactive 3D digital twin with zero floating-point rotational drift.
 - Maintain zero cloud dependencies: 100% local processing.
 - Provide accessible, responsive interfaces across desktop, tablet, and mobile browsers.
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7. Non-Goals (V1)

- Physical motor or servo control (ESP32 / Arduino hardware).
 - User authentication, accounts, or persistent cloud databases.
 - Multi-puzzle support (4x4, 2x2, Pyraminx).
 - Automated unguided free-form 3D object tracking without reticle alignment.
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8. Scope Categorization (MoSCoW)

MUST HAVE (V1)

- WebRTC camera feed with rear/environment camera preference.
- 3x3 square reticle target with perspective-safe ROI.
- Trimmed-mean pixel aggregation and HSV/CIELAB color classification.
- Temporal stability buffer requiring consecutive stable frames before capture.
- Guided 6-face capture sequence (U, F, R, B, L, D) with physical rotation hints.
- Strict state validation: 54 stickers, 9 of each color, 6 canonical centers, corner twist parity, edge flip parity, and permutation parity.
- Client-side Herbert Kociemba two-phase solver.
- Interactive 3D Rubik's cube with 27 cubies.
- Zero-drift layer rotation animations with orthogonal matrix snapping.
- 3D directional curved move arrows.
- Full playback controls (Play, Pause, Step Next, Step Previous, Reset to Scramble, Speed selector).
- Real-time Computer Vision Debugger inspection panel.
- Fallback simulated scramble demo for camera-less environments.

SHOULD HAVE (V1.x)

- Manual sticker color correction override prior to solving.
- Audio cues for frame capture and move execution.
- Auto-capture toggle settings in preferences.

COULD HAVE (V2)

- Real-time move verification via live webcam stream.
- Web Worker multithreading for table generation.

OUT OF SCOPE (V1)

- FUTURE / OUT OF SCOPE FOR V1: ESP32 microcontrollers, stepper motors, servo drivers, robotic grippers, and physical chassis.

9. User Stories

- US-01 (Scanning): As a user, I want the camera viewfinder to guide me through holding each face in a clear sequence so that all 54 stickers are accurately digitized.
- US-02 (Stability): As a user, I want a visual progress bar indicating when the cube is held steady enough to be captured so that motion blur does not cause misclassifications.
- US-03 (Validation): As a user, I want instant validation that alerts me if a piece is physically impossible so that I can rescan only the problematic face without starting over.
- US-04 (Solving): As a user, I want an efficient move sequence generated in milliseconds without having my camera data uploaded to any server.
- US-05 (3D Guidance): As a user, I want to see the digital twin rotate the exact layers with 3D directional arrows so that I know exactly which face and direction to turn on my physical cube.
- US-06 (Playback): As a user, I want to step forward, backward, pause, or auto-play through the solution at varying speeds.

10. User Journey

```
journey
  title Cubyntra User Journey
  section Initialization
    Open Cubyntra Web App: 5: User
    Grant Camera Permission: 4: User
  section Scanning
    Align White Face: 5: User, Camera
    Hold Still (Stability Lock): 5: Vision
    Follow Physical Rotation Cues (6 Faces): 5: User, Vision
  section Processing & Solving
    Validate Physical State & Parities: 5: Engine
    Run Kociemba Two-Phase Solver: 5: Solver
  section 3D Guidance
    Inspect 3D Scrambled Digital Twin: 5: User, Three.js
    Step Through Moves with 3D Directional Arrows: 5: User, Three.js
    Execute Solution to Solved State: 5: User, Confetti
```

11. Functional Requirements

- FR-CAM-001: Application shall request camera permissions with rear/environment preference.
- FR-CV-001: Application shall sample 9 distinct sticker sub-regions inside a defined central ROI.
- FR-CV-002: Application shall aggregate pixel values using trimmed mean to reject specular glare.
- FR-CV-003: Application shall classify colors into White, Yellow, Green, Blue, Red, Orange.
- FR-CV-004: Application shall require consecutive frame consensus before triggering stability lock.
- FR-CUBE-001: Application shall maintain an independent logical model of 54 facelets and 27 cubie coordinates.
- FR-CUBE-002: Application shall validate sticker counts, centers, edge flip parity, corner twist parity, and permutation parity.
- FR-SOLVER-001: Application shall execute Herbert Kociemba's two-phase algorithm client-side.
- FR-SOLVER-002: Application shall return Half Turn Metric (HTM) sequences verified against an internal simulation.
- FR-3D-001: Application shall construct an interactive 3D Rubik's cube twin.
- FR-3D-002: Application shall snap cubie positions and quaternions to orthogonal integer axes upon rotation completion.
- FR-3D-003: Application shall render 3D curved directional arrows pointing in turn directions.

12. Non-Functional Requirements

- NFR-PERF-001: Layer animation execution shall run at a consistent 60 FPS on supported hardware.
- NFR-PERF-002: Kociemba two-phase solve search shall return solutions in under 100ms on typical scrambles.
- NFR-PRIV-001: Zero image or video data shall leave the client device.
- NFR-A11Y-001: All controls shall be keyboard-navigable and respect prefers-reduced-motion.
- NFR-REL-001: Application shall gracefully handle camera denial or absence via synthetic scramble fallbacks.

13. Acceptance Criteria

- When 6 valid faces are scanned or mock loaded, the solver generates a valid solution sequence.
- Applying the solution sequence to the 3D cube results in all 6 monochrome faces.
- Arbitrary rotations produce exactly 0.000% floating point drift.
- An intentionally twisted corner or flipped edge produces a clear, actionable validation error.

14. Risks & Mitigations

Risk	Severity	Mitigation
Heavy ambient lighting cast (yellow incandescent)	Medium	Center sticker white-balance calibration and CIELAB perceptual metrics
Motion blur during hand movement	High	Temporal stability buffer requiring consecutive identical frames
Browser camera permission denial	Medium	Informative UI with instant simulated scramble demo button
Floating-point matrix accumulation drift	High	Integer matrix and position snapping after every layer turn

15. Future Robotics Expansion

FUTURE / OUT OF SCOPE FOR V1

Cubyntra's logical move abstraction (CubeMove[]) is designed to decouple algorithm generation from physical execution. In future versions (V3/V4), the same move array can be routed to an ESP32 hardware motor controller driving 6 stepper motors or a 2-gripper mechanism to physically solve cubes automatically.